

Sarah G. Kane

Cambridge, UK — sgk27@cam.ac.uk

EDUCATION

University of Cambridge, Cambridge, UK
PhD in Astronomy

October 2023 — Present
Supervisors: Vasily Belokurov and Stephanie Monty

University of Pennsylvania, Philadelphia, Pennsylvania
B.A. in Physics with Distinction

August 2019 — May 2023
GPA: 4.0 — *summa cum laude*

EXPERIENCE

Research Assistant

University of Pennsylvania, Department of Physics and Astronomy May 2020 – May 2022, May 2023 - August 2023

Senior Honors Thesis: “Chemical Classification of Gaia Spectra Using Machine Learning”

Department of Physics and Astronomy, University of Pennsylvania September 2022 – May 2023

NSF Research Experience for Undergraduates (REU): “Machine Learning with Gaia DR3”

Department of Astronomy, University of Texas at Austin May 2022 – August 2022

HONORS AND AWARDS

Marshall Scholarship	2023 - 2026
Institute of Astronomy Community Service Award for Public Outreach	2026
Dean’s Scholar, University of Pennsylvania	2023
The LGBT Alumni Association Student Leadership Award, University of Pennsylvania	2023
Roy and Diana Vagelos Science Challenge Award, University of Pennsylvania	2022 - 2023
Michael McGowan Leadership Scholarship, National Organization for Albinism and Hypopigmentation	2022
Thomas H. Wood Award, University of Pennsylvania Department of Physics and Astronomy	2020

PUBLICATIONS

You can find a full list of my first and co-author publications in this ADS library.

First & Co-First Author Publications

Kane, S. G.[†], Kaczmarek, Z.[†], Garner, A., Buder, S., & Monty, S., (2026). “A value-added catalogue of neural network-based Europium abundances for GALAH DR4,” *MNRAS*, 547, 2, 10.1093/mnras/stag209. ^{†these authors contributed equally to this work}

Kane, S. G., Belokurov, V., Monty, S., Baumgardt, H., Filion, C., Kravtsov, A., Myeong, G., & Zhang, H. (2026). “On the connection between nitrogen-enhanced field stars and the Galactic globular clusters,” *MNRAS*, 546, 1, 10.1093/mnras/staf2272.

Kane, S. G., Belokurov, V., Cranmer, M., Monty, S., Zhang, H., & Ardern-Arentsen, A. (2025). “The ones that got away: chemical tagging of globular cluster-origin stars with *Gaia* BP/RP spectra,” *MNRAS*, 536, 3, 10.1093/mnras/stae2689.

Arcand, K. K.[†], Schonhut-Stasik, J. S.[†], Kane, S. G.[†], Sturdevant, G.[†], Russo, M., Watze, M., Hsu, B., & Smith, L. F. (2024). “A Universe of Sound: Processing NASA Data into Sonifications to Explore Participant Response,” *Frontiers in Communication*, 10.3389/fcomm.2024.1288896. ^{†these authors contributed equally to this work}

Co-Author Publications

Shepherd, R. D., Harrison, C. M., Caselton, G. et al. (incl. Kane, S. G.) (2026). “Evaluating Sonification for Effective Learning and Enjoyable Experiences in Astronomy,” (*submitted*).

Naidu, R. P., Oesch, P. A., Brammer, G., et al. (incl. Kane, S. G.) (2026). “A Cosmic Miracle: A Remarkably Luminous Galaxy at $z_{\text{spec}} = 14.44$ Confirmed with JWST,” *OJAp*, 9, 10.48550/arXiv.2505.11263.

Zhang, H. Y., Belokurov, V., Sanders, J. L., et al. (incl. Kane, S. G.) (2025). “Orbital migration and heating history of the Galactic disc: a transition between the bimodal discs,” (*submitted*).

Davies, E. Y., Belokurov, V., Kravtsov, A., et al. (incl. Kane, S. G.) (2025). “Blind source separation of the stellar halo,” *MNRAS*, 539, 2, 10.1093/mnras/staf558.

Ardern-Arentsen, A., Kane, S. G., Belokurov, V., Matsuno, T., Montelius, M., Monty, S., & Sanders, J. L. (2025). “Predicting metallicities and carbon abundances from *Gaia* XP spectra for (carbon-enhanced) metal-poor stars,” *MNRAS*, 537, 2, 10.1093/mnras/staf096.

Zhang, H., Belokurov, V., Evans, N. W., et al. (incl. Kane, S. G.) (2025). “Observational Constraints of Radial Migration in the Galactic Disk Driven by the Slowing Bar,” *ApJL*, 983, L10, 10.3847/2041-8213/adc261.

Zhang, H., Belokurov, V., Evans, N. W., **Kane, S. G.**, Sanders, J. L. (2024). “Kinematics and dynamics of the Galactic bar revealed by Gaia long-period variables,” *MNRAS*, 533, 3,10.1093/mnras/stae2023.

SUCCESSFUL OBSERVING PROPOSALS

- PI: 10 hours on VLT/UVES for observations of europium-enhanced stars in the Galactic disk 2026
- CoI: 19 hours on VLT/UVES for high precision spectroscopy of the globular cluster NGC 6397 2026
- CoI: 31 hours on VLT/UVES for follow-up of carbon-enhanced metal poor stars from *Gaia* BP/RP spectra 2025
- CoI: 17 hours total on VLT/UVES for follow-up of metal-poor, α -poor stars from BP/RP spectra 2025 - 2026

TALKS

* denotes an invited talk

Seminars

- *Stellar Populations Journal Club, Liverpool John Moores University March 2026
- *Astrophysics Seminar, University of Surrey January 2026
- *Astronomy Journal Club Talk, Rutgers University [*virtual*] December 2025
- *Astronomy Lunch Talk, University of Pennsylvania October 2025
- Tea Time Talk, Geneva Observatory March 2025
- Wednesday Seminar, Institute of Astronomy, University of Cambridge November 2024
- *Astronomy Lunch Seminar, Newcastle University May 2024

Conferences

- *APOGEE's Galactic Odyssey: Reflecting on the APOGEE Survey and Steve Majewski's Contributions Teruel, Spain August 2026
- Ordo ab chao*: Star formation and chemistry from $z > 10$ to local relics Kavli Science Focus Meeting, Cambridge, UK February 2026
- Bridging scales: star clusters and their host galaxies from the Local to the high-z Universe Matera, Italy September 2025
- The Role of Star Clusters in Star Formation from Local to Galaxy Scales National Astronomy Meeting, Durham, UK July 2025
- Unseen Astronomy: Multi-sensory approaches for research, communication and education National Astronomy Meeting, Durham, UK July 2025
- *A Universe of Sound: Astrophysical Data Sonification Workshop, Harvard & Smithsonian CfA October 2024
- Connecting the Milky Way with its parent populations at the dawn of time, National Astronomy Meeting, Hull, UK July 2024
- Designing for diversity in astronomy and geophysics, National Astronomy Meeting, Hull, UK July 2024
- * Kafka & Disability Series (*panel*), Bodleian Library, Oxford University, UK June 2024
- 241st meeting of the American Astronomical Society (*poster*) January 2023

TEACHING

Students Supervised

- **Christine Han**, Master's (Part III) Astrophysics, University of Cambridge 2024 - 2025
- **William Ju**, high school student, now undergrad at Cornell University 2023
- **Daniel**, high school student (surname omitted for privacy) 2023
- **Arianna Ridgeway**, high school student, now undergrad at Wellesley College 2023
- **Divya Karmani**, high school student, now undergrad at the University of Pennsylvania 2021 - 2022

Courses

- Guest lecturer, **ASTR401: Introduction to Astronomy**, New Mexico State University 2026
- Co-instructor, **Nonvisual Data Science Workshop Series**, *introductory Python tutorials designed for blind and low vision learners, supported by a grant from Pandas and NumFOCUS* 2024
- Co-instructor, **Undergraduate Summer Computing Training**, *computing courses offered to summer research students at the Department of Physics and Astronomy at the University of Pennsylvania* 2023

PROFESSIONAL ACTIVITIES

Conference Organizing

- SOC, “Sonification for Research: Techniques, Efficacy, and Applications (SoniTEA),” Flatiron Institute Center for Computational Astrophysics October 2025
- SOC, “Unseen Astronomy: Multi-sensory approaches for research, communication and education,” UK National Astronomy Meeting July 2025

Service

- Referee for A&A, ApJS
- Chair, EDI Committee Inclusion and Fairness Focus Group, Institute of Astronomy 2025 - Present
- Co-chair, LGBTQ+ Focus Group, Institute of Astronomy 2025 - Present
- International Women’s Day Committee, IoA, Cambridge 2024
- Science Olympiad Urban Initiative at the University of Pennsylvania Co-chair 2022 - 2023; Volunteer 2019 - 2023

SELECTED OUTREACH AND PRESS

Advocacy

- Statement to the UN Human Rights Council on behalf of the Hermansky-Pudlak Syndrome (HPS) Network and in collaboration with the International Centre for Trade and Development March 2026
- UN Human Rights Engagement Training; UN Office of the High Commissioner for Human Rights and UN Independent Expert on the enjoyment of human rights by persons with albinism 2026
- Equity and Inclusion Journal Club, Rutgers University October 2025
- Statement to the UN Human Rights Council (on behalf of the US National Organization for Albinism and Hypopigmentation (NOAH) and the Hermansky-Pudlak Syndrome (HPS) Network and in collaboration with the World Organization Against Torture) March 2025
- EDI Seminar, Institute of Astronomy, University of Cambridge March 2025
- Invited panelist, “Listen to the Universe with the Chandra X-Ray Observatory,” *US Library of Congress* February 2025
- Invited Panelist, “Sonification: Why Multi-modal Science is the Future,” *arXiv Accessibility Forum 2024* September 2024
- Invited Panelist, “Whose responsibility is it to make content accessible?” *arXiv Accessibility Forum 2023* April 2023

Outreach and Community Engagement

- “Europium: the coolest element you’ve never heard of,” public talk at the Institute of Astronomy, University of Cambridge December 2025
- Institute of Astronomy Open Day, Cambridge 2024 - 2025
- Subject Matter Expert, NASA Universe of Learning 2024 - 2025
- “Are white dwarfs stars?: A debate,” public talk with Katherine Kauma at the Institute of Astronomy, University of Cambridge March 2025
- “Hearing the Unseen Universe,” talk at the Institute of Astronomy, University of Cambridge February 2024
- Invited Panelist, “Sonification and the Future of Generative Music,” *SXSW* March 2023
- Invited Panelist, Women in STEM Symposium at the University of Pennsylvania March 2023
- Co-founder, STEM Connections, National Organization for Albinism and Hypopigmentation 2021 - 2023
- Organizer, field trip for students from Overbrook School for the Blind to the University of Pennsylvania’s Department of Physics and Astronomy December 2021

Press

- Featured in “How sounds can turn us on to the wonders of the universe”, *MIT Technology Review* June 2023
- Invited guest, “Episode 153 - Let’s Meet Sarah Kane” [Podcast], *Limitless: The Blind Beginnings Podcast* 2023
- “Who, What, Why: Sarah Kane finds ways to represent data as sound”, *Penn Today* March 2023
- Featured in “Five Trending Topics We Saw at SXSW 2023”, *Ad Council* March 2023
- Featured in “Audio Astronomy Unlocks a Universe of Sound”, *Scientific American* January 2023
- Invited guest, “Episode 60: An Ear for Education (Sonification 2)” [Podcast], *Astro[sound]bites* August 2022
- “Disability Advocacy and the Sciences”, *Penn OMNIA* January 2022